

SUMMER RESEARCH PLAN

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I am planning for a research visit beginning from 18/05/2026 until 03/07/2026. During this period I will be visiting some of my collaborators, for example Pinakinath Saha (assistant professor at IIT Delhi), Krishna Hanumanthu, (professor at Chennai Mathematical Institute and also my PhD advisor) and Bivas Khan (assistant professor at IIT BHU). We have an ongoing research collaboration (details later) which we want to advance and make some progress.

Background: Let X be an algebraic variety and L be a line bundle on X . A line bundle L is said to be *very ample* if the rational map $\phi_L : X \rightarrow \mathbb{P}^n$ defined by the global sections of L is an immersion for some n , and in addition $\phi_L^*(\mathcal{O}_{\mathbb{P}^n}(1)) = L$ [9].

A Line bundle L is then said to be *ample* if $L^{\otimes n}$ for some $n \gg 0$ is very ample [9]. Equivalently, Nakai-Moishezon-Kleiman criteria for the ampleness of a line bundle says:

Definition 0.1 (Nakai-Moishezon-Kleiman [11]). *A line bundle L is ample iff $L^{\dim V} \cdot V > 0$ for every irreducible subvariety $V \subset X$.*

Similarly, one can define the numerical effectiveness (nefness) of a line bundle L on X as follows:

Definition 0.2 (Nef [11]). *A line bundle L is numerically effective (nef) iff $L^{\dim V} \cdot V \geq 0$ for every irreducible subvariety $V \subset X$.*

The following is a characterization for ampleness of a line bundle L on a projective variety X to be ample.

Theorem 0.3 (Seshadri's criterion for ampleness [8]). *Let X be a projective variety and L be a line bundle on X . Then, L is ample if and only if for every $x \in X$, there exists a positive number $\varepsilon > 0$ such that*

$$\frac{L \cdot C}{\text{mult}_x C} \geq \varepsilon$$

where x is an arbitrary point of X and $C \subset X$ is any reduced and irreducible curve containing x .

Using the above characterization of ampleness of a line bundle, Demailly defined the Seshadri constant of a line bundle at a point [7].

Definition 0.4 (Seshadri constant at a point). *Let X be a smooth projective variety and L be a nef line bundle on X . Let $x \in X$ be a point, then the Seshadri constant of L at x is defined as*

$$\varepsilon(X, L, x) := \frac{L \cdot C}{\text{mult}_x C}.$$

where the infimum is taken over all reduced and irreducible curves $C \subset X$ passing through x .

Taking a further infimum and supremum over all points we get the following two constants:

Definition 0.5. (a) $\varepsilon(X, L) := \inf_{x \in X} \varepsilon(X, L, x)$.
 (b) $\varepsilon(X, L, 1) := \sup_{x \in X} \varepsilon(X, L, x)$.

Note that the Seshadri constant $\varepsilon(X, L, x)$ is a real number which lie in the interval $[0, \sqrt[n]{L^n}]$, where L^n is n -th self intersection of L . In-fact for an ample line bundle L we have the following sequence of inequalities:

$$0 < \varepsilon(X, L) \leq \varepsilon(X, L, x) \leq \varepsilon(X, L, 1) \leq \sqrt[n]{L^n}.$$

Ampleness of vector bundle: The notion of ampleness and nefness is extended to vector bundles in the following natural way. Let X be a projective variety and E be a vector bundle on X , then E is said to be ample (nef) if $\mathcal{O}_{\mathbb{P}(E)}(1)$ is ample (resp. nef) on $\mathbb{P}(E)$.

The guiding research problems in the area of positivity are as follows:

- (i) Determine when a given line bundle L or a vector bundle E on a projective variety X is ample (nef).
- (ii) Given an ample (nef) line bundle L or a vector bundle E on a projective variety X , compute the Seshadri constant of L or of E at a point $x \in X$.
- (iii) If computing $\varepsilon(X, L, x)$ is difficult, provide a bound.
- (iv) Determine if $\varepsilon(X, L)$ is a rational number.

Objective 1: Let X be a normal projective variety equipped with an action of a semisimple algebraic group G , and assume that X contains a unique closed orbit. Let B be a Borel subgroup of G and let E be a B -equivariant vector bundle on X . For more details about notations and definition see [1], [10], and [6]. In this setting, we would like to explore the following

Question 0.6. *If the restriction of E to every curve in X is ample, then under what conditions is E itself ample?*

This question has been studied in several situations. An affirmative answer is given when E is a torus-equivariant vector bundle, either on a toric variety (see [?, Theorem 2.1]) or on a generalized flag variety (see [4]). A related question was studied for equivariant vector bundles on wonderful compactifications in [5], for Bott-Samelson-Demazure-Hansen varieties or wonderful compactifications of a symmetric varieties of minimal rank in [3], and for G -Bott-Samelson-Demazure-Hansen varieties in [2].

Objective 2: Additionally, we are interested in computing the Seshadri constants of ample line bundles and vector bundles on simple G -projective variety X . In particular, we are interested in computing

- (1) $\varepsilon(X, L; x)$, for an ample line bundle L at a point $x \in X$
- (2) $\varepsilon(X, E; x)$, for a B -equivariant nef vector bundle E at $x \in X$

Objective 3: Let X be a compact Riemann surface and (E, ϕ) be a pair, where E is a vector bundle on X and ϕ is a section of E , i.e., $\phi \in H^0(X, E)$. We then consider $X \times \mathbb{P}^1$ and the two projections $q_1 : X \times \mathbb{P}^1 \rightarrow X$ and $q_2 : X \times \mathbb{P}^1 \rightarrow \mathbb{P}^1$. We get the following short exact sequence of bundles:

$$0 \rightarrow q_1^* E \rightarrow W \rightarrow q_2^*(\mathcal{O}(2)) \rightarrow 0.$$

We would like to express the ampleness/nefness of W in terms of the pair (E, ϕ) .

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